

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

This assessment features a quick guide to the non-verbal reasoning tests, as well as an introduction to the shape questions your child might encounter. Then, there are two practice tests for your child to have a go at. We have included an answer guide too so that you can help your child to identify any misconceptions.

What skills does this practise?

Non-Verbal Reasoning

Problem-Solving

Spatial Reasoning

Shapes

Further Activity Ideas and Suggestions

For more non-verbal reasoning practice, check out this [11+ Non-Verbal Reasoning Analogies Assessment Pack](#) and this [11+ Non-Verbal Reasoning Week of Workouts](#). You can encourage your child to get in some bite-size practice of all the 11+ skills with this set of [11+ Activity Mats](#).

Parents Blog



Twinkl Kids' TV



Homework Help



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Parents Hub

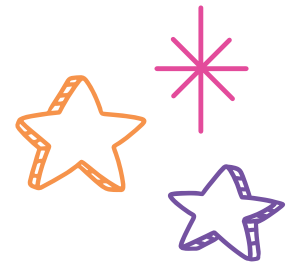


Non-Verbal Reasoning

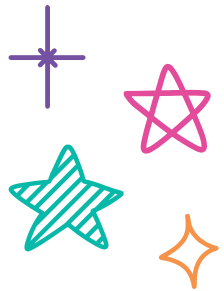


The non-verbal reasoning test involves solving problems that are represented with pictures and diagrams. There are a number of different types of questions for the non-verbal reasoning test which could include:

- finding shapes that are like each other based on similarities (for example colours and number of sides);
- finding the odd one out in a group of pictures;
- identifying a shape which has been rotated;
- breaking the code in a sequence of images;
- spotting the missing shape in a sequence;
- identifying the nets of 3D shapes.



Here is an example of the type of question you may get in this section:



1. Which is the correct code for figure 4?

 M B	 R B	 M Q	 ?
1	2	3	4

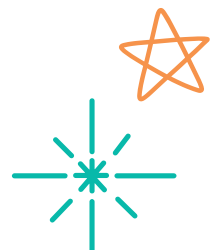
B	Q	B	R
Y	M	Q	Q
A	B	C	D



The answer would be D. The letter common to the first two shapes is B which represents the cross-hatching pattern they both have. Therefore Q represents the solid orange colour. The M is the letter representing the shape of the moon and it comes at the top of the box. Therefore the answer is D (R to represent the irregular dodecagon and the Q at the bottom to represent the solid orange colour).

The non-verbal reasoning test, for many children, can be very intimidating. The best way to prepare for this is through practice. Complete many different type of practice non-verbal reasoning tests with your child and discuss how each question is solved. This will give your child the experience needed to tackle the on-verbal reasoning tests.

For a thorough guide to all aspects of the 11+ test, try our **Survival Guide for Parents/Carers.**



11+ Non-Verbal Reasoning Practice Paper

Shape

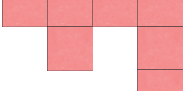
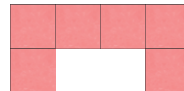
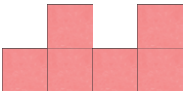
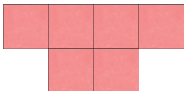
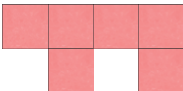
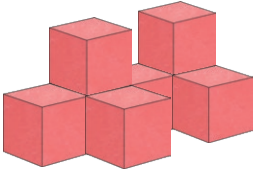
A common type of question that appears on many non-verbal reasoning papers is the shape question.

This question type requires you to find the correct net to match a given 3D shape. There are a variety of different question types that you may come across, such as finding the correct net to match a top-down view of a 3D shape (focusing on the layout of the shape) or finding the correct net to match the sides of a 3D shape (focusing on the pattern that is on the sides).

You will usually be provided with a 3D shape to look at and then a selection of nets to choose from. Alternatively, you may be given a net to look at and a selection of 3D shapes to choose from.

Here is an example of a top-down question:

Tick the figure that shows what this shape would look like as viewed from above.

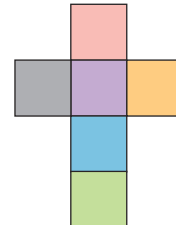
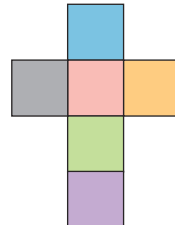
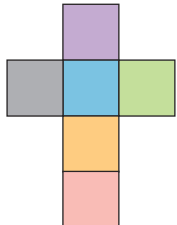
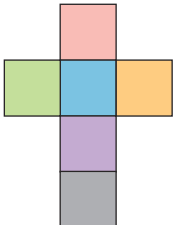
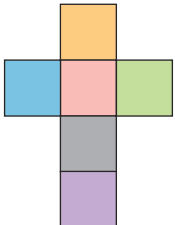
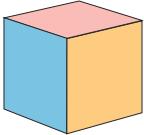


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The answer is the first figure.

Here is an example of a 3D net question:

Tick the net that matches the 3D shape.



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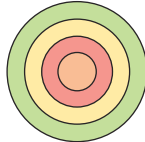
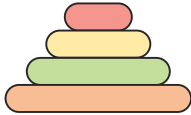
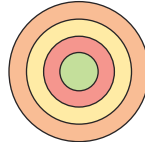
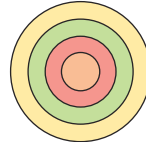
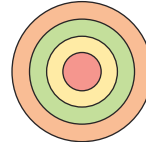
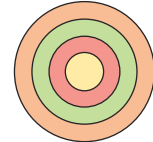
The answer is the second figure.

11+ Non-Verbal Reasoning Practice Paper

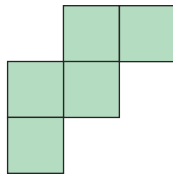
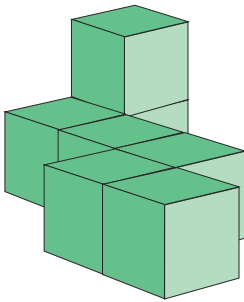
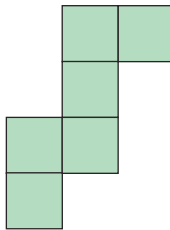
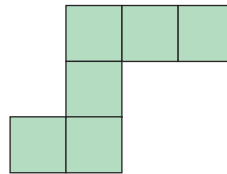
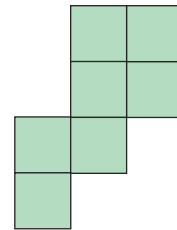
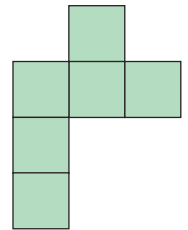
Shape: Test 1

Tick the figure that shows what the shape would look like as viewed from above.

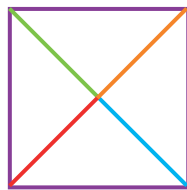
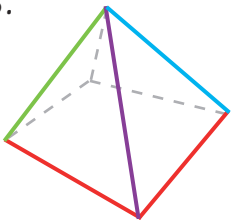
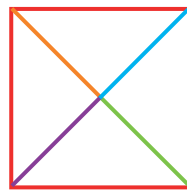
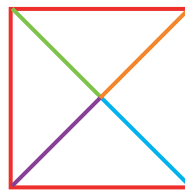
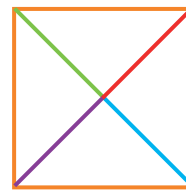
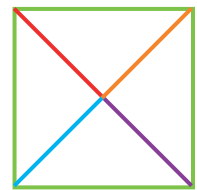
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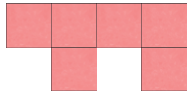
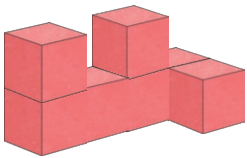
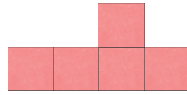
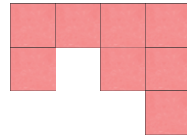
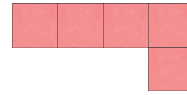
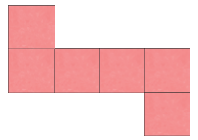
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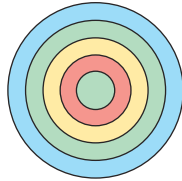
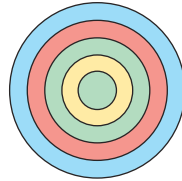
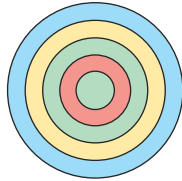
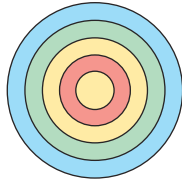
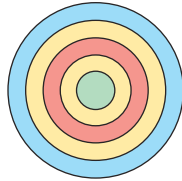
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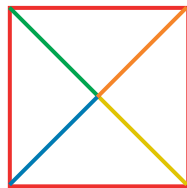
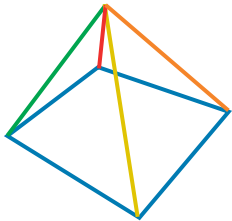
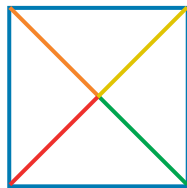
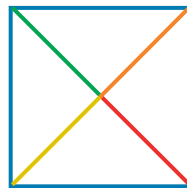
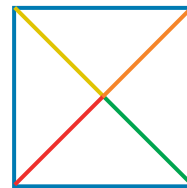
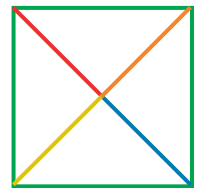
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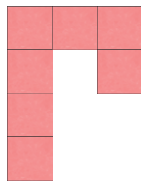
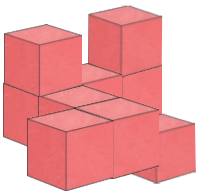
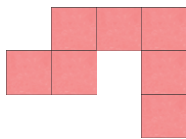
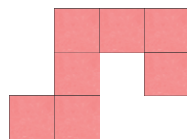
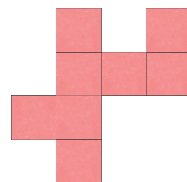
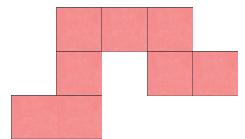
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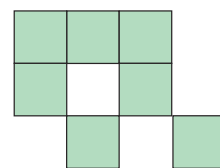
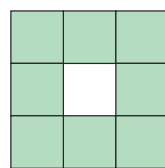
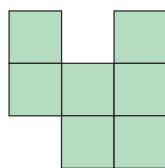
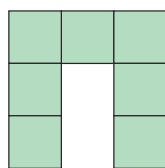
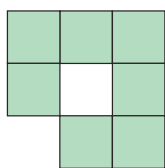
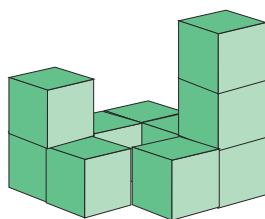
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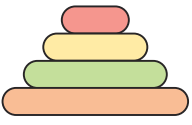
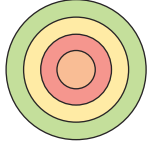
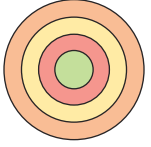
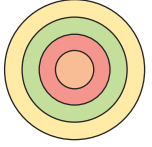
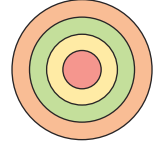
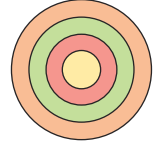


11+ Non-Verbal Reasoning Practice Paper:

Shape: Test 1 Answers

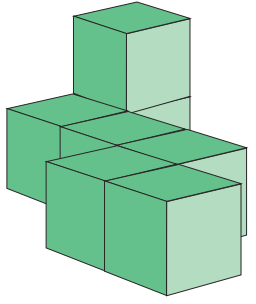
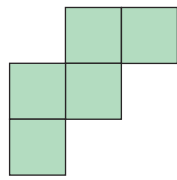
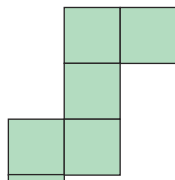
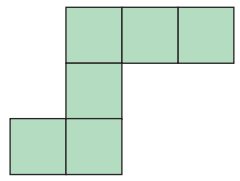
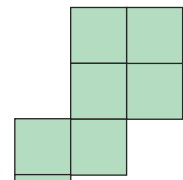
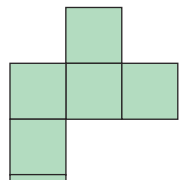
Tick the figure that shows what the shape would look like as viewed from above.

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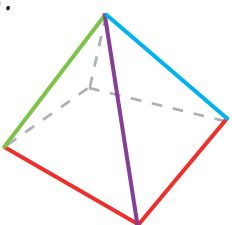
The largest disc is the orange one, followed by green, yellow and red.

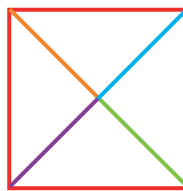
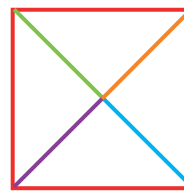
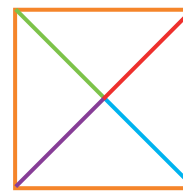
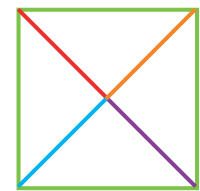
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We can see that there are six surfaces visible on the top of the shape. This means it could only be options 2, 3 or 5. We can also see that the shape has one square further forward than the others. This eliminates options 3 and 5, meaning it must be option 2.

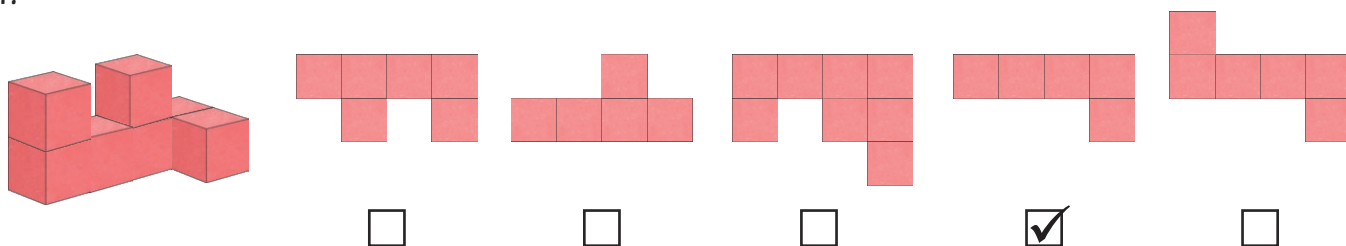
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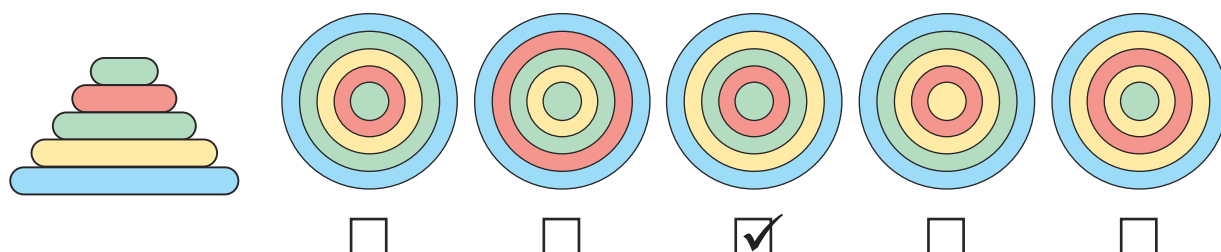
The base of the pyramid is red, meaning that the square needs to be red. This means it can only be option 2 or 3. The purple edge is between the green edge and the blue edge. This means it must be option 3.

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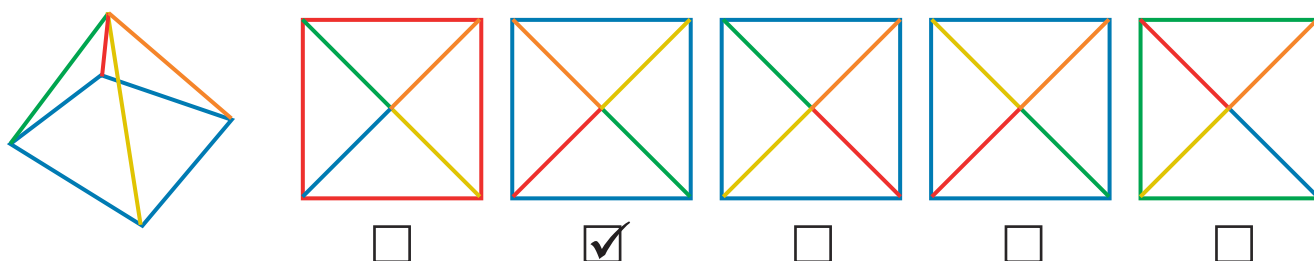
Looking at the top surfaces of the shape, we can see that our shape has 5 squares, meaning it must be option 2 or option 4. The shape also has a single cube slightly further forward, at the very end of the shape. That means it must be option 4.

5.



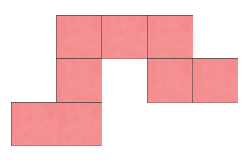
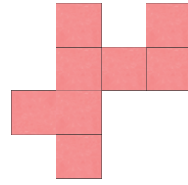
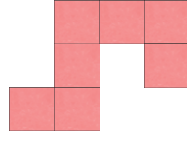
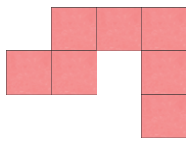
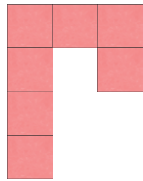
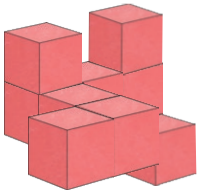
The largest disc is blue, followed by yellow, green, red and green.

6.



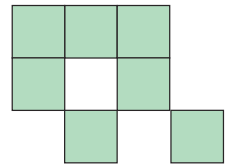
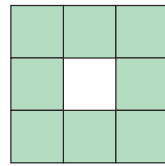
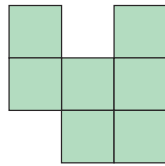
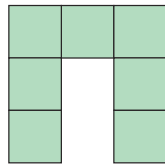
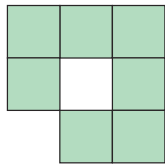
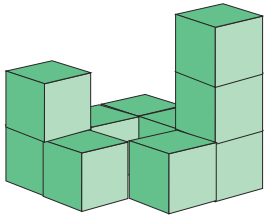
The base of the pyramid is blue, meaning it must be option 2, 3 or 4. The edges are yellow, green, red and orange, with red and yellow opposite each other and green and orange opposite each other. This means it must be option 2.

7.



The shape contains seven surfaces visible on the top, meaning we need seven squares. This means it must be option 1, 2 or 3. There is a cube at the front and to the left of the shape, meaning it can't be option 1. This cube protrudes 2 squares in front and one to the left of the back row, meaning it must be option 3.

8.



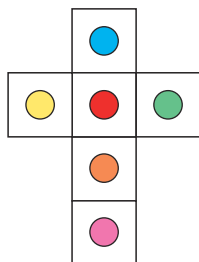
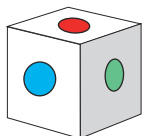
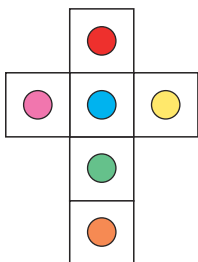
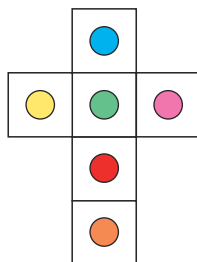
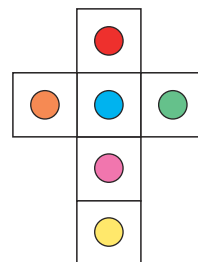
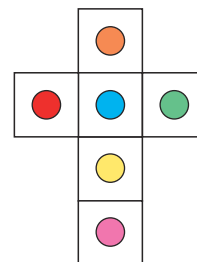
There are seven surfaces visible on top of the shape, meaning it cannot be option 4. There is a cube missing from the front left corner, meaning it cannot be option 2. There is no cube to the right of the shape, meaning it cannot be option 5. There is a cube missing from the centre, meaning it cannot be option 3. The answer is option 1.

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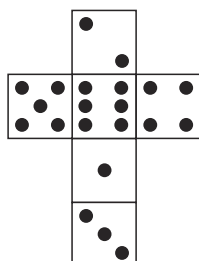
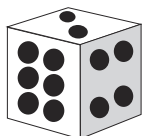
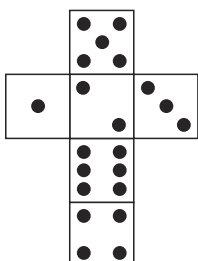
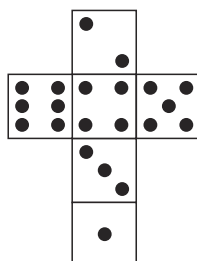
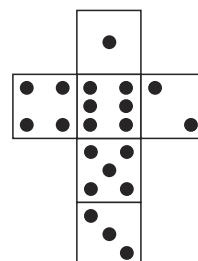
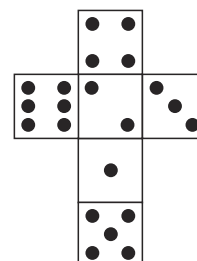
Shape: Test 2

Tick the net that matches the 3D shape.

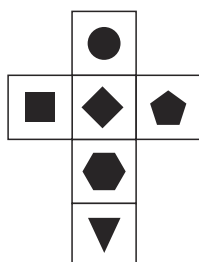
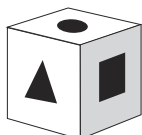
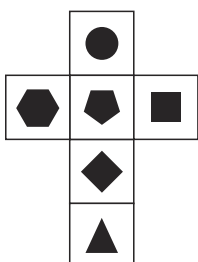
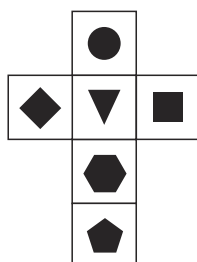
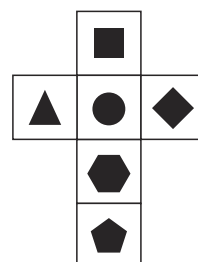
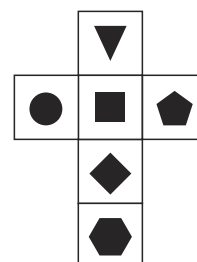
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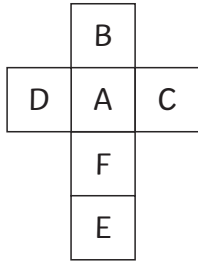
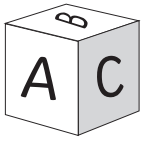
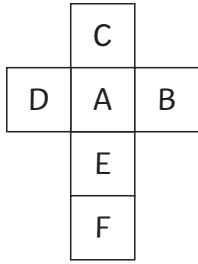
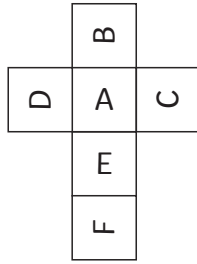
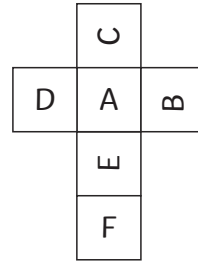
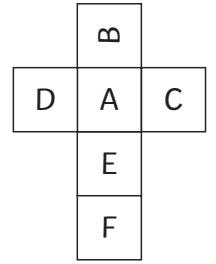
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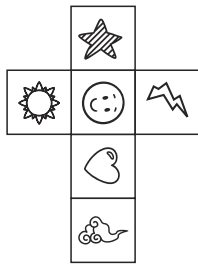
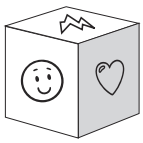
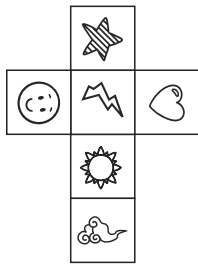
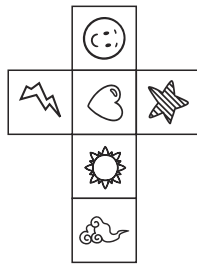
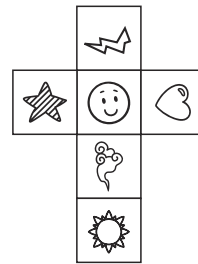
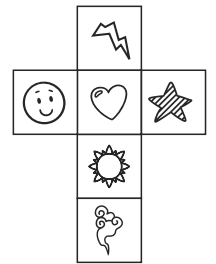
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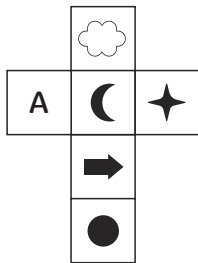
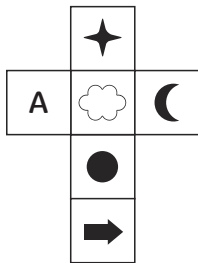
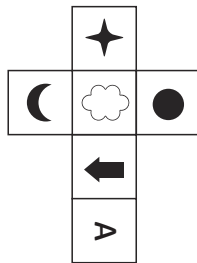
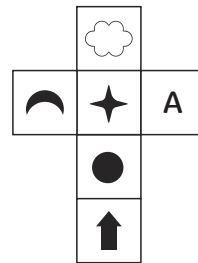
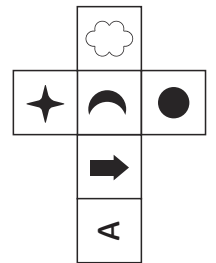
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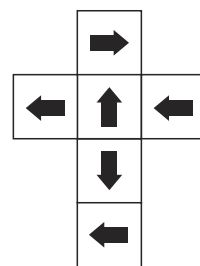
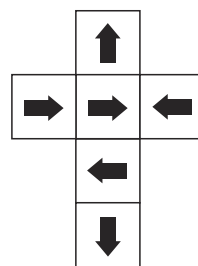
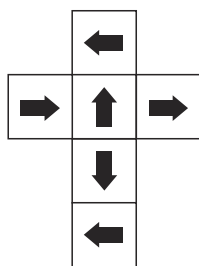
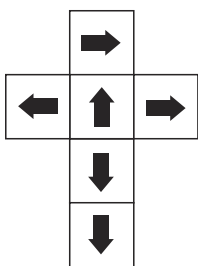
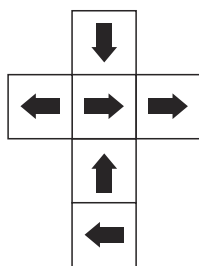
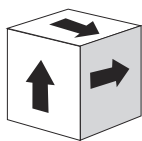
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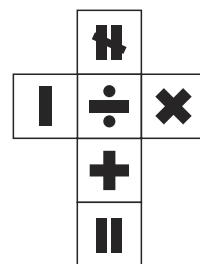
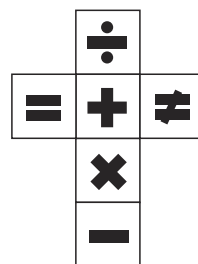
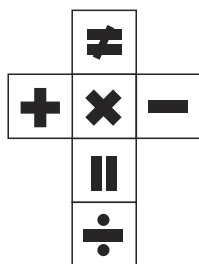
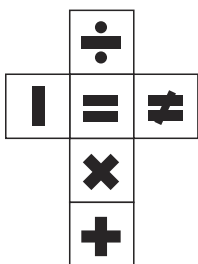
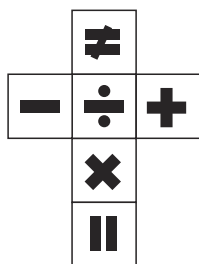
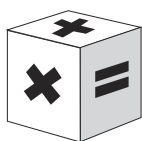
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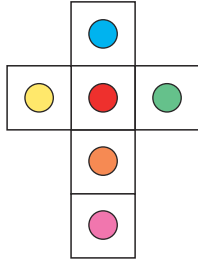
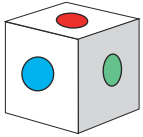
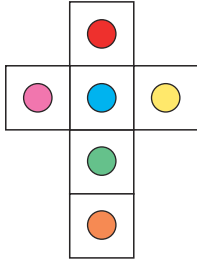
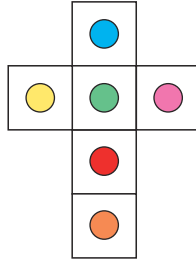
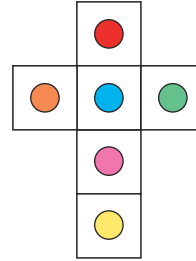
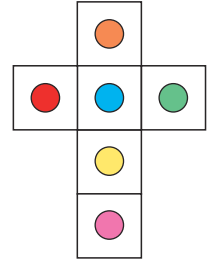


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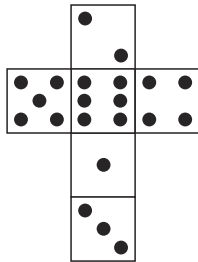
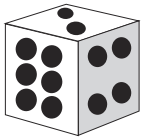
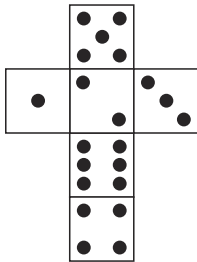
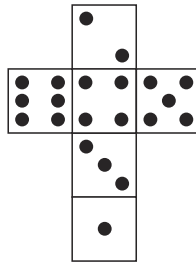
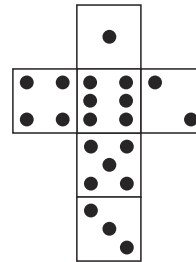
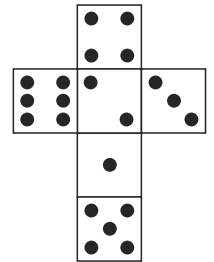
Shape: Test 2 Answers

Tick the net that matches the 3D shape.

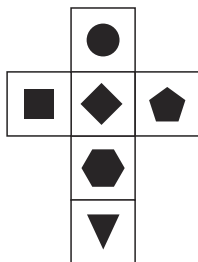
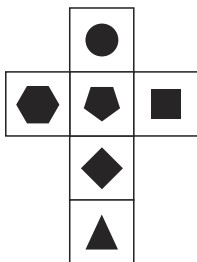
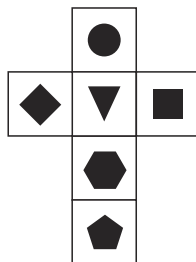
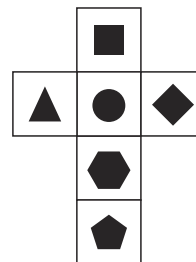
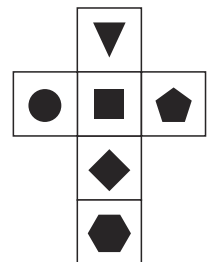
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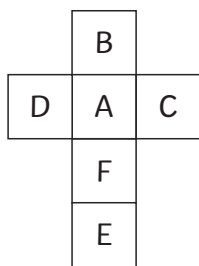
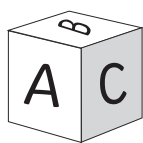
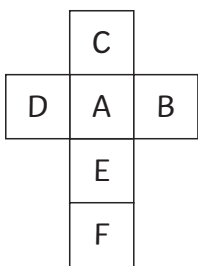
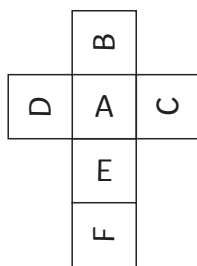
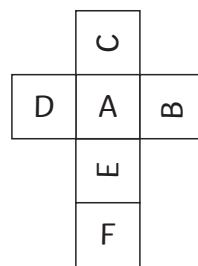
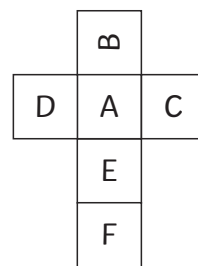
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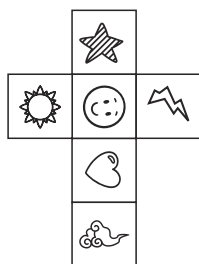
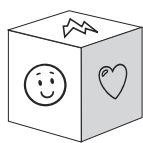
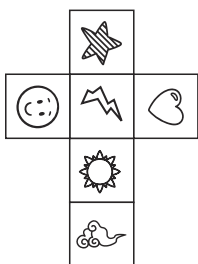
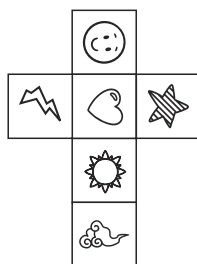
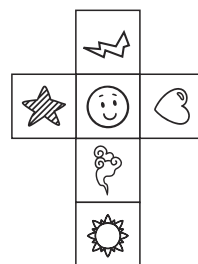
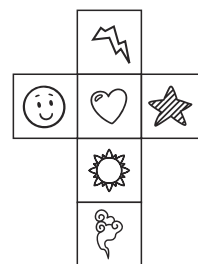
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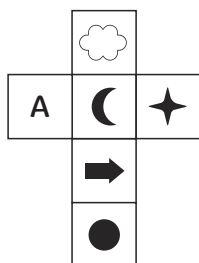
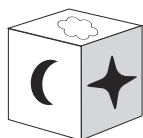
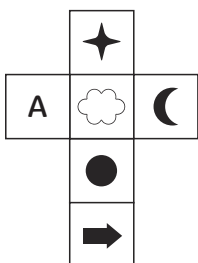
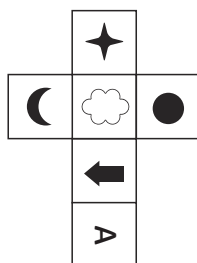
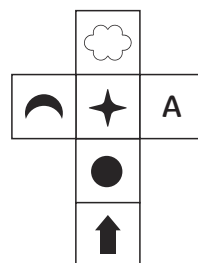
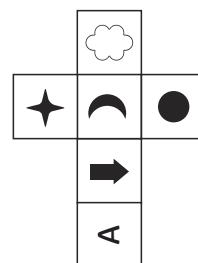
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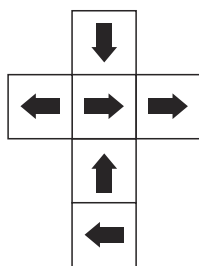
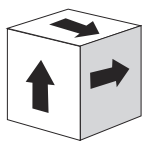
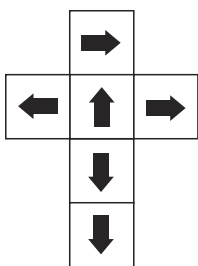
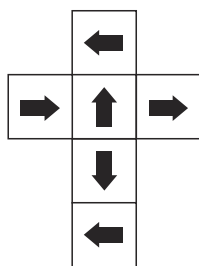
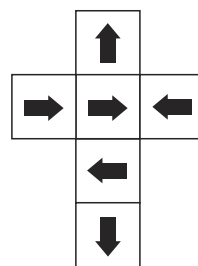
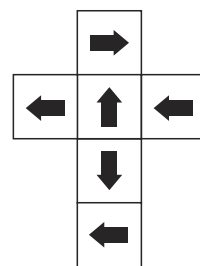
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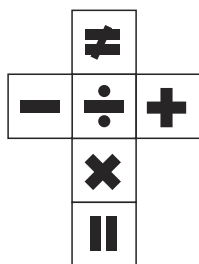
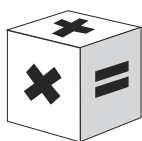
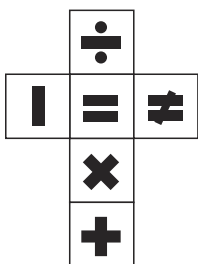
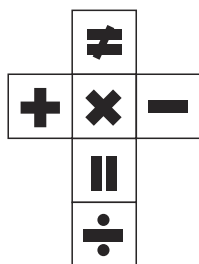
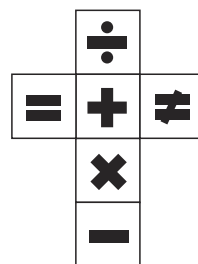
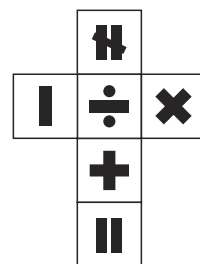
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